CSC 163: Computing, Ethics, and Society
ABET SO4 · Module A3 (4c)

Data Privacy Methods Exercises Solution

Department of Computer Science,
Hofstra University

Exercise: Threat to method

Choose the best primary protection method for each threat: encryption in transit, encryption at rest, access control, data minimization, k-anonymity, differential privacy.

Threat	Method
Reading traffic on public Wi-Fi	
A lost laptop with the customer database	
A support agent browsing every customer record	
Storing birth dates only to send a newsletter	
Releasing hospital records to researchers	
Publishing usage statistics so no single user can be inferred	

Exercise Solution: Threat to method

Match the method to where the risk comes from: outsiders, insiders, over-collection, linking, or inference from statistics.

Threat	Method
Reading traffic on public Wi-Fi	**Encryption in transit**
A lost laptop with the customer database	**Encryption at rest**
A support agent browsing every customer record	**Access control (least privilege, logging)**
Storing birth dates only to send a newsletter	**Data minimization**
Releasing hospital records to researchers	**k-anonymity**
Publishing usage statistics so no single user can be inferred	**Differential privacy**

Exercise: Compute k

A released table has the groups below (same ZIP prefix and age band = one group). (a) What is k? (b) Which group is the weakest? (c) Give two ways to reach k of at least 3. (d) What weakness remains even if the table is k-anonymous?

Group	ZIP	Age band	Records
A	130xx	under 30	3
B	130xx	30 or over	2
C	148xx	under 30	4
D	148xx	30 or over	5

Exercise Solution: Compute k

(a) k is the size of the smallest group: **k = 2**. (b) Group B (130xx, 30 or over).

(c) **Suppress** the 2 records of group B, so the smallest group is 3 (k = 3); or **generalize** age to 'any' for ZIP 130xx, merging A and B into 5 records, so k = 4.

(d) **Homogeneity:** if every record in a group has the same sensitive value (for example, the same diagnosis), the attacker learns it anyway.

Group	ZIP	Age band	Records
A	130xx	under 30	3
B	130xx	30 or over	2
C	148xx	under 30	4
D	148xx	30 or over	5

Exercise: Randomized response

Each person flips a coin: heads, answer truthfully; tails, flip again and answer yes on heads, no on tails. So P(yes) = 0.5 p + 0.25, where p is the true fraction of yes. Find p from each observed result.

Observed yes	Computation	True p
40%	2 × (0.40 − 0.25)	
55%	2 × (0.55 − 0.25)	
30%	2 × (0.30 − 0.25)	
25%	2 × (0.25 − 0.25)	

Exercise Solution: Randomized response

Solve p = 2 × (observed − 0.25). Check: p = 0.60 gives 0.5 × 0.60 + 0.25 = 0.55. Why is privacy kept? No one can tell whether an individual answer was true, but the group statistic is recoverable.

Observed yes	Computation	True p
40%	2 × (0.40 − 0.25)	**30%**
55%	2 × (0.55 − 0.25)	**60%**
30%	2 × (0.30 − 0.25)	**10%**
25%	2 × (0.25 − 0.25)	**0%**

Exercise: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
Removing names from a dataset makes it anonymous		
Pseudonymized data is no longer personal data under GDPR		
Encrypting a database at rest stops a legitimate key holder from misusing it		
In differential privacy, a smaller epsilon means stronger privacy		
Data minimization means collecting everything now and filtering later		
With public-key encryption, the sender uses the recipient's public key		

Exercise Solution: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
Removing names from a dataset makes it anonymous	F	Quasi-identifiers (ZIP, birth date, gender) can re-identify people
Pseudonymized data is no longer personal data under GDPR	F	It is reversible, so it is still personal data
Encrypting a database at rest stops a legitimate key holder from misusing it	F	Encryption does not limit those who hold the key; use access control and logging
In differential privacy, a smaller epsilon means stronger privacy	T	—
Data minimization means collecting everything now and filtering later	F	Collect only what the purpose needs and delete it afterwards
With public-key encryption, the sender uses the recipient's public key	T	—

Exercise: Practice 4c question

A university wants to release course-enrollment data to researchers. Describe **two methods** to protect student privacy, explain **how each works and its limits**, and **compare** them. (Privacy Report style: about one page.)

Exercise Solution: Practice 4c question

- **Model answer.**
- k-anonymity: generalize ZIP and age so every combination appears at least k times; this blocks simple linking. Limits: homogeneity and background-knowledge attacks, and less detail.
- Differential privacy: release only statistics (counts per course) with calibrated noise, so no one student changes the answer much. Limit: noise hurts small groups; a privacy budget limits repeated queries.
- Comparison and recommendation: k-anonymity keeps row-level data for researchers but is weaker; differential privacy gives a provable guarantee for statistics but not row-level detail. Recommend differential privacy for published tables, k-anonymity with access agreements for shared records.
- **Scoring (4 points):** two correct methods (1); how each works (1); limits of each (1); comparison and reasoned recommendation (1).